CSS BHAM

Computer Science Module Choice Opinions For 2025-26

What is this?

This document is a place for people to comment on modules running next year, so younger students can get advice from older students about what modules will suit you best. Think of it as a mini-Wikipedia for modules next year, but opinions as well as facts. Feel free to put whatever you want below, and also edit other pieces of advice to neaten things up. If you don't agree - don't just delete. Comment underneath saying you disagree or explain why.

Disclaimer: This is just crowdsourced so take the information with a pinch of salt; look at the department's actual comms and website about stuff for more information.

This document is historic - if you want to give feedback for modules from academic year 25-26 please go here:
Module Choice Opinions For 2026-27

Links!

- https://docs.google.com/document/d/1-bzpzRw1sSFtvBOnycmuuAnsqWfzYs_dJ8_g-rBwxh4 (last year's doc)
- Full List of Programmes (Select your year to see the options you have!)
- CSS Discord (Feel free to ask anyone questions about the modules here!)

Key

Code & Link	Module Titles	Semester 1 Credits	Semester 2 Credits
• Opinions			

Compulsory Modules

06-26581	**BSc Computer Science Project**	**20**	**20**

- Try and keep on top of this for as much of semester one as you can, especially if you're planning on doing a 3:1 split, it's very easy to fall behind.
- For some reasons, supervisors seem to really love saying "don't worry about writing up until you're finished/after your demo". I would not trust that. It was so fucking stressful.
- Try to finish most of your project during the first semester. Since exams have been moved to the summer, I highly recommend that you try to complete as much project work as possible over the winter break.
  Writing will be hard, literally broke me and caused burnouts at some points. As long as you know the ins and outs of your project, the development, testing and evaluation section of the writing will be trivial; the hardest sections are introduction and literature review.
  Try to get as many meetings with your supervisor as possible for feedback, and get other people to read your writing, as it really helps with spotting mistakes.

06-26587	**MSci Computer Science Project**	**20**	**40**

- Start thinking about the structure of your report early, and write it as you go.
- If you're told to pivot your project 6 weeks into semester 2, good luck!
- Contrary to what Matt Leeke told us, and has likely told you, it **is** important to find a good supervisor with an interest in the field that you wish to do a project in. Or at least it was for me.

Optional Modules

06-40087	**Advanced Cryptography**	**-**	**20**

- Absolutely goated module. It's difficult but achievable - it won't be *easy* by any means, but is taught well enough that everything is *understandable*. Also, it is a lot of fun 😀
- This module really impressed me with how well organised everything was, considering it's an entirely new module. Elisabeth is a fantastic lecturer, and super supportive if you have any questions or need any help. The content is definitely challenging, some weeks you may need to set aside a good bit of time to understand wtf is going on, but once it makes sense it's really fun. The coursework is two 25% assignments, and they're pretty well structured. Overall, a really good module, I recommend it if Cryptography is your thing, particularly if you enjoyed the public key cryptography bit of S+N.
- This year the coursework assignments have been set up to give you as much chance as possible to get the full grade so with a good bit of work, they are very achievable.
- Second assignment cooked a lot of people

06-40087	**Advanced Cryptography**	-	**20**

- But also, if you looked at the distribution of marks people got, on average it was almost full marks, even though it was very tricky.
- This module after reading week made me question my life choices
  - Edit: I take it all back, 10/10 module
- Probably the best module in all of compsci. Super well organised and well paced. Had time to learn even more content honestly. Assignments were structured to allow you to get full marks even without getting everything right. Focus on actually understanding the content and this was reflected even in the exam.

06-35309 06-35325	**Advanced Functional Programming** **Advanced Functional Programming (Extended)**	-	**20**

- hmmmmmmmmmm. I am conflicted about this module. On the one hand - it's super interesting, and unlike anything I've ever seen. It's all about proving correctness of functions and building things and encoding behaviours and properties from first principles (arguably it's not what I would think of when I think "programming" but whatever) and if that sounds difficult you are **correct**. It's a challenge in so many ways, and while the majority of the module team is great, I'm not sure it's the best quality module I've ever taken. There's a lot of info on the GitLab repository, but it's not the best organised and often it's just like "this thing has these properties and it looks like this and works like this" and spews literal screen-loads of Agda at you with no further explanation. The lectures are also fairly intense - they're *all* live coding and if you miss something, it's gone, AND some of the lecturers give teeny tiny breaks (5 minutes at best) - so you've got to be fairly well focused all the time (not something I excel at). Right now I'm very behind and I'm not quite sure how I'm going to catch up.
  - Even if you think you can hack AFP, it will torture you :)
- This year the module is taught by three lecturers - Martin, Eric and Todd. They rotate eac weed h week (I think?) and teach roughly the same way, mainly live coding. This can vary in quality - some work by following along others just write as they go which can be hard to follow. I have found the content quite satisfying, but requiring a fair amount of work outside of lectures to get a hang of using agda and being familiar with Types and proofs that are predefined.
- Not for the faint of heart. Can't fault the teaching of the module at all, but this module is a serious time commitment to get a reasonably decent grade (see: the avg grade in the practice test).
- I agree this is not an easy module, but I think Martin, Todd, and Eric are all fantastic at what they do (A 10/10 module for me, only beat by 2nd year FP). Going to labs and lectures is a necessity, and office hours are a great way to get that extra help from the lecturers & TAs. The content covers content from MLFCS, ToC, FP, DSA, and other Theory modules throughout the course, as well as building up new ideas in Agda.
- Agda is a very different approach to programming than Haskell is (Haskell is more general purpose, Agda is a theorem proving language) and so even if you enjoyed

06-35309 06-35325	**Advanced Functional Programming** **Advanced Functional Programming (Extended)**	-	20

FP with Haskell, you may find Agda is not for you because though the syntax is similar, the aim of programming (and even the approach with all the C-c C-, etc., shortcuts) is another world entirely.

- I wanted more Functional Programming so I chose this module. I did not get more Functional Programming. This module's name is false advertising, should be called Advanced Logical Foundations of Computer Science or something instead.
- This module is not easy, but it is very fun. The lab and lectures are basically mandatory (you will struggle even more if you don't show up to them) and there are a lot of office hours which are very helpful in helping you understand what the fuck is going on. The lecturers are also really nice and helpful although as said above this is not functional programming 2. I would only recommend this module if you enjoyed things like mlfcs, toc and (if you take it) plpdi
- Overall I'm really glad I took this module, I really really enjoyed it once everything started clicking in my head, and the fact that it's 100% coursework is a god-send once it gets to exam season. That being said, there were some weeks in the middle of term I was working crazy hours to be able to actually understand what was going on and not fail the exams, which come up quicker than you'd expect! We're very lucky to have such accomplished lecturers in the field, but they can be a little bit in the sauce… Again though, would reaffirm, really enjoyable & an awesome introduction to type theory, proof assistants, and the more logical world of CS.
-

06-30209 06-38963	**Advanced Networking** **Advanced Networking (Extended)**	20	-

- Storytime with Ian
- His lectures are really good but sometimes it's unclear what you should really be making a note of.
- Personally, I've found the content dry (I didn't plan on picking this module though, it replaced CAV not running this year) and so it's tricky to feel motivated outside the lectures - I think it's a little overrated, but that's my opinion
- Assignments aren't particularly tricky, but I found them hard to motivate myself to start them
- I somewhat expected the content to be dry but felt it would be good knowledge to have. I took this instead of neural computation as a last minute switch because I heard the teaching there was mid. The lectures are wonderful because Ian is a phenomenal lecturer. Assignments felt a bit tedious to start but they were alright in the end. Personally I think this module was great overall though I have bias towards good lecturers as opposed to a more objective view of the content (though I'm pretty sure the content was good, both thorough and relevant).
- Good module, Ian is a brilliant lecturer and managed to keep me awake in the 9am lectures. Content is a bit dry but Ian genuinely makes up for it. He is also really

06-30209 06-38963	**Advanced Networking** **Advanced Networking (Extended)**	**20**	**-**

relaxed with assignments, didn't care if you uploaded it late and they were generally pretty easy albeit a bit boring

- Ian is good at explaining things in detail. I actually found the long tangents really useful for putting the lecture content in context. I actually found most of the module content very interesting since everything presented was always clearly linked back to real world problems. It's not very clear what we need to know for the exam as there are many tangents, examples and tips that seem superfluous, but nevertheless very interesting and likely to prove useful throughout our careers.
- Worth the 9ams - engaging and interesting, and Ian's always happy to chat about the content afterwards if you have questions. Can be a challenge to sift through all the extraneous waffle but he's also fairly chill about it all - the assignments are less "I'm looking for this set thing" and more "I'm looking to see that you understand where this all fits together and have the ability to think", plus he appears not to believe on late penalties and just says "submit it late if you need but don't take the piss". Big props, might be my favourite module to date.
- Pretty good module and Ian is nice, but there's an awful lot of content.
- Agree with the comment on making the 9ams worth it. I found the lectures pretty engaging. Oh and don't do crimes.
- Ian's a really good lecturer and knows his stuff if you go to his office hours. How interesting you find the content will likely depend on your opinion of networking in general (duh). There's two assignments: one is worth 5% of the module and you just have to get wireshark working (easy to get 100%), the other is worth 15% and you have to do some networky stuff in wireshark, it's relatively straightforward. Ian will let you submit late as long as you don't take the piss which is very helpful when you have other assignments in the last week or two. Only downside is that there is a lot of content and it's not clear what needs to be learnt and what is just there for historical context. Overall, I really enjoyed this module.

06-35308 06-35326	**Algorithms and Complexity** **Algorithms and Complexity (Extended)**	**-**	**20**

- **Love** this module. It's a solid progression from DSA and (to a lesser extent) ToC but you're not going to be stuck if you don't remember much from them. Various different examples of algorithms for different problems, analysis of their functioning, correctness and runtime and a fairly large amount of "here, take this algo and modify it to do something else". Rajesh is a great lecturer, explains things well, goes at a really comfortable pace and is super approachable with any questions (plus he has a lovely voice). The weekly assignments are good for keeping you in lock-step with the content, too. And the slides are good. Solid recommendation.
    - As a follow up, post-exam: The exam! The exam was good!
    - Came out of it feeling like my brain had been exercised but not faced with anything that felt impossible, just a nice challenge - much like the

06-35308 06-35326	**Algorithms and Complexity** **Algorithms and Complexity (Extended)**	-	**20**

assignments and rest of this module.
    - That's to say, the exam was much like the rest of the module - sound.
- Really interesting and well taught module, the lecturer is friendly and amazing at explaining hard concepts. I would recommend this module.
- Once again, you need to keep up with weekly work.
- Underrated module, was considering dropping out at first as the lecturer made it seem like it would be hard, but it isn't. Go over the weekly content in your own time, especially because you have weekly assignments to do. If you don't understand something, just message Rajesh on Teams; he replies really quickly.
- For those wondering, there is no programming in this module.
- *Seriously, 30 marks for one 3-hitting set question?? Rajesh u ok?*

06-30211 06-30239	**Computer-Aided Verification** **Computer-Aided Verification (Extended)**	**20**	-

- RIP (no longer running) (good riddance)

06-32167 06-32212	**Neural Computation** **Neural Computation (Extended)**	**20**	**-**

- Concepts were explained clearly and if you have a general interest in AI I would actually recommend this module. The slides contain all the information that is required for the assessments without any ambiguity and are well annotated. Of the two lecturers, Alexander Krull is far far better and you can tell he has a genuine passion for teaching the topic and tries to emphasise the harder concepts and break them down for you. Some of the content is challenging but I believe the slides do a good job of explaining them clearly. I would highly recommend using YouTube videos by 3Blue1Brown to clarify some of the foundational concepts
    - I second the 3B1B recommendation!!
- If I had any other decent option available to me I would have left this module behind in Week 1. Genuinely up there as one of the worst modules I've done. The coursework section was two 10 question quizzes that weren't too difficult if you'd properly gone over the content, and were open book, so that was pretty fair. But dear god, motivating myself to go to these lectures was like pulling teeth. They were dry as all hell and had my brain melting out of my ears with boredom. I would have rather done CVI again. As someone who is lukewarm on AI, pick a different module and save your brain.
- If you want to understand new AI topics like transformers and how ChatGPT works, this is a great course. The topics covered include RNNs, autoencoders, diffusion models, GANs, and transformers. I personally didn't like the Edgbaston lectures as they were difficult to understand, so I watched the Dubai recordings instead, he explains everything really clearly (Kashif Rajpoot), he's the best. Each week, there's a lab file in Jupyter Notebook. Some quiz questions are based on these labs. Most of the code is already written, but there are a few lines left for you to complete nothing too complicated, however the labs could be very long. There are two online quizzes with 10 multiple-choice questions. They aren't too hard if you've studied well, but make sure to read the questions carefully. If this dubai lecturer (Kashif) was not there, I would've had a completely different opinion about this module.
- Incredible for the new AI topics and implementing them in the weekly labs. Not as theoretical as ML but more practical. Do this if you actually want to learn the real architecture behind ai instead being another "prompt engineer"
- Best module, I highly recommend this module before you actually take classes such as Computer Vision or NLP. It would give you a general understanding in those field before you actually go into them. Especially if you have a lot of modules in your hands this one is a must.
- most interesting module i took in first semester. lectures were really really helpful and alex explained things really well, not just reading off the slides. the quizzes are challenging but if you stay up to date with the work you'll be fine. very fun module, highly recommend. currently my favourite module so far

06-30213 06-30241	**Computer Vision and Imaging** **Computer Vision and Imaging (Extended)**	**20**	**-**

- This module ***SUCKS*** do not pick it. The lecture material (slides) are unhelpful - they're long, but most slides have no actual words on them. Unless you can fully understand every concept B.O. from pictures alone you cannot learn the content just from the slides. The lectures themselves are also unhelpful - most of the time rather than actually explaining the content they say something along the lines of "the data goes into the filters where convolution happens and then you get an output". That means nothing. The worst part of the module is the assignment worth 50% of the entire module. Firstly, there is no detailed description of how we're marked, which is just unhelpful. Then there's a 3 page limit, which is incredibly strict for no reason, especially when we're expected to show our results (as in, a lot of images). There's also a 5 page limit for "supplemental material" as in code, but this isn't marked so I don't know the point of it. Lastly, they were not clear as to if we're allowed to use premade libraries like skimage to do the edge detection for us. One person says we're expected to use stuff we learnt in the labs (which includes the libraries), then another person says we should implement it ourself, then that person backtracks to saying using the libraries is ok. In conclusion the 50% coursework was awful because it wasn't clear in what it wants, and the 50% exam at the end of the year is going to be awful because both ways of learning the content are unclear.
- I am genuinely dreading having to somehow find a way to "revise" this module from the notes I have made on it over the past semester. About 80% of my notes are just screenshots of images that literally mean nothing. o7 glhf
- First half of the module is okay and the lab work is interesting, after week 5 it turns into a slew of images shat all over the PowerPoint in no coherent order and without any meaning. The slides may as well be AI generated. As mentioned by somebody else, the assignment is ridiculous and has the most nonsensical rules for something that makes up 50% of our final grade. I dread having to find the willpower to translate this sea of images into some form of knowledge that can be used to study for the exam.
- The lecturers changed the rules about the assignment at least 3 times - one of those being less than 24 hours before the deadline, only take this if you're prepared to resubmit your work 8 times and cry over trying to get all your screenshots into a word doc.
- If you want to take this module I would give you a few tips to help you. Personally for me I enjoyed the module slightly…. HOWEVER, like mentioned. There are no words in the lecture slides and mostly pictures so you do have to watch the lecture videos. I would recommend you to use an effective note taking system to help you. As you listen to the lecture videos, write down 80-90% of what they are saying. I would recommend doing this digitally to help you out. Use Google docs or Notion to take your notes. Have a structured way to do this. The module is not the best structured in terms of slides etc so you have to really try your best and make it work for you.
- Look at the past papers and see how questions are asked so when you are studying for your exams you know what you should focus on. As in the lectures they can go off on a tangent and just talk it about everything but nothing in particular.

06-30213 06-30241	**Computer Vision and Imaging** **Computer Vision and Imaging (Extended)**	**20**	**-**

Have a positive mindset towards this module if you really want to take it. I kind of had to take it because there was no other module I could take personally as I didn't have interest in any other module. So I was determined to “make it work” for me.

- The assignment was not too hard but instructions were NOT clear at all. TAs told us one thing while the lecturers were saying another thing which just made everything a bit wishy washy. The task itself was not hard but I do wish the instructions were clearer and there were heavy markers for stuff. We had a limited amount of space to write due to the 3 page limit. I managed to squeeze everything in but i could not go in depth in regards to some stuff because of it
- I appreciate that the lectures were recorded as trying to revise using the slides on their own is pretty impossible. If you are not mathematically inclined like me, then you may also benefit from going over the recorded lectures slowly, instead of attending the in-person lectures as I felt quite overwhelmed with the pace and lack of clear explanation for quite a few topics.
- As aforementioned the coursework was pretty vague and poorly managed, but if you can get past that then it was pretty easy.,
- One of the best AI modules I’ve done. Content is clear and really well laid out (you have to watch the videos / attend lectures as the slides are not as useful alone). The coursework at first was confusing in terms of what they are asking us to do, however when you realise that implementing means do it yourself without the use of packages then it’s pretty chill. How to implement methods taught is clearly explained with algorithms + explanation in the slides / lectures and labs. Three page limit is a bit annoying but quite easy to condense if you have references and start early. Highly recommend reducing line spacing to give yourself more room as they don’t penalise for it. Exam felt very fair and with the 50/50 split was a lot less stressful. Revise all content as they can and do examine nearly all weeks. If they focus on explaining algorithms further or provide slides specific to calculation / application for parts of content then it will likely come up in the exam. 8/10 module, clear lectures, fairly straightforward coursework and the fairest exam I’ve done after 4 years of university. Highly recommend it if you're interested in CV.

06-38963 06-38966	**Dependable and Distributed Systems** **Dependable and Distributed Systems (Extended)**	**-**	**20**

- It’s the Matt Leeke module, anxntent is pretty good and interesting, but the teaching style is mid. Matt spends a lot of time waffling and not teaching - half the lecture will be spent recapping the previous lectures, the other half will be spent going through the content at faster than comfortable speeds because there’s suddenly not enough time to get through it and half of that time is peppered with stories. He also seems to view lectures as a bit of a game? He’ll try and do things to give lectures extra value… like not recording them. His argument is that recording a lecture diminishes the value of it because it encourages people not to come, so he doesn’t provide any lecture recordings, has pre-recordings on Canvas (which, in all fairness, I’m told are

06-38963	**Dependable and Distributed Systems**	-	**20**
06-38966	**Dependable and Distributed Systems (Extended)**		

ok) and then includes random extra bits and pieces in the lectures like stuff about exam questions and the coursework which, in principle is great, but feels like you're being toyed with in actuality ("this week's added value for the 9am is going to be me telling you what ⅓ of your test is!" and it turns out to be him saying "it'll be something from the first third of the module!")

- I second what the person above me said, and I also want to add: the coursework is a lit review worth 20% (correct me if I'm wrong), get started on it earlier and make sure to get feedback. Also, I learned way too much about Matt during the lectures, including stuff I really did not care about, which would be fine if it was a quick comment here and there, but he would go on tangents (and go on them often!) about topics that had absolutely nothing to do with the content, which got annoying after a few lectures. The content was super interesting in however, and doing the lit review is good practice for FYP if you're doing an MSci.
- It's a good module if you want reassurance that you have a lecturer who knows what they're talking about, won't try to be mean-spirited or catch you out - Matt's feedback on the lit reviews was genuinely useful to me. Like mentioned previously, the content and presentation is a bit dry though, so it's sometimes difficult to get through (coming from someone who does have an interest in DDS). But at least I know Matt isn't actively trying to fuck me over like with Mobile
- It is quite an easy module, there isn't much to learn and Matt is more than happy to explain things further if you don't understand them. The coursework is very helpful for your FYP and you get feedback on it if you send it to him beforehand. As for his lectures, I enjoy the 5-10 minutes of him just talking about things at the start and makes me more likely to show up, although sometimes he does go through things a bit too fast, but he does that because he goes through the 'less important' things very quickly, and also hints towards what is likely to show up on the exam, which you don't get if you don't go to the lectures, although it is entirely possible to learn everything from the pre recorded lectures, he does give good reason to show up to the lectures and not many modules can say that. He also responds to emails quicker than he probably should so any questions you have are responded to very quickly.
- be me
  get feedback
  "operating at an excellent level"
  "you can expect a high first"
  feel pretty good about myself
  implement all feedback given
  get mid 70s
  guy I know who just GPTed his entire paper and submitted it in 2 hours
  gets 80
  mfw
- I actually disagree that Matt's unrelated yaps encourages me to attend lectures, I had the opposite experience. My attendance to lectures plummeted once I realised there's only about 25/30 mins of Matt actually saying something useful. Obv

06-38963 06-38966	**Dependable and Distributed Systems** **Dependable and Distributed Systems (Extended)**	-	**20**

subjective but I wasn't a huge fan of the lit review assignment. Also agree with above regarding Matt's lecture style- not terrible by any means just meh. Personally, I found the content to be incredibly boring. If you're unsure if you'd find it interesting or not, check out Nancy Lynch's Distributed Algorithms (1996) and/or similar textbooks. That being said, there's not a whole lot of content and it's very doable.

- FUCK THIS SHIT
  - FUCK THIS SHIT
- It was a part of Advanced Networking and it should have stayed that way. The lectures are a waste of your time if you don't think somebody yapping about Taylor Swift, his Apple stocks or Dark Souls is funny; the coursework had barely any real guidance given unless you emailed him and the content is so boring that I fell asleep multiple times revising it. Adding two more whole sections to the module (Dependability Concepts and the Broadcast Algorithms / Clocks) didn't help at all.
- The content on this module might be the most boring thing i have ever looked at in my life. Im fighting myself to not fall asleep while i read about whatever random bullshit he yapped about, and its basically impossible to get more than 80% unless if you've written a phd level paper for both the coursework and the exam.
- I regret the choices that led me to revising this module
- I don't know how everyone else thinks that other lecturers make any module more interesting than it is, Matt Leeke is one of the only lecturers that seems like he knows how boring the content is so he tries to make it more tolerable, boohoo he talks about something unrelated for 5 fucking minutes. There is a LOT of content and it is all broad concepts so an understanding of one thing will only help you for that one thing, there is not a lot of flow/carry over content so you will have to revise for a long time. To get the best grade, I'd advise you to do anki cards from day 1.

06-35310 06-35376	**Evolutionary Computation** **Evolutionary Computation (Extended)**	-	**20**

- The content is very interesting if you ever liked those youtube videos where a population evolves and reproduces (What are Genetic Algorithms?)
- Courseworks can be time consuming but is gratifying as you improve and iterate on your algorithm to perform better. I genuinely felt like I learnt something in this module instead of being taught the same content but in different flavour for the tenth time.

06-40088 06-40093	**Game Theory** **Game Theory (Extended)**	-	**20**

- Quite a lot of maths, only do it if you like maths! The lecturer likes to draw links from other random fields so it could be a pro or con depending on your range of interests.

06-40089 06-40094	**High Performance Computing** **High Performance Computing (Extended)**	**20**	**-**

- Ben is sound, the module is genuinely really interesting and cool, the slides are really nicely laid out and easy to understand, the 20% coursework was actually really fun - we were essentially given a relatively simple program to write, then had to experiment with different combinations of ways to optimise it and record the results. Took a punt on this module despite it being new this year out of curiosity and honestly I'm really glad I did. Shoutout again to Ben, honestly he made this module just by being real cool and always bringing the energy and the vibes.
- Amazing module, take it if you want no AI and like C programming. But there is maths; calculus
- Incredibly well put together given the fact it was a new module. This is a must pick if you like low level programming and squishing all the performance out of the hardware. Actually quite beneficial for getting into developing the backend framework of AI/ML. The lecturer has a background in physics so it's a must pick if you like coding simulations like fluid, thermal or mechanical.
- My favourite module, very good for people who are interested in hardware and and distributed processing
- RIP x2

06-30214 06-30251	**Human-Computer Interaction** **Human-Computer Interaction (Extended)**	**20**	**-**

- 100% Group project, is basically the design part of Team project where you just come up with an idea and justify your research and design points. Great it you have a group that all shows up, you can get away with only putting in an hour a week if everyone stays on top of the work otherwise it could all pile up at the end. Support sessions if you need your slides checked. Easy module as long as you dont forget about the coursework
- The Cheese module, it's really easy but requires a lot of time commitment due to it being 100 percent coursework and a team based module.
- Easy but boring and there are people who got not amazing grades for the module(below 70%) . Personally would never do it again if i had the choice but thats because apparently im bad at creating a slide show
- I would recommend it if you don't mind teamwork and just want an easy module. It's not mind numbingly boring like other easy modules either (personally). Exactly as the 1st point said.
- Very much a cheese module in terms of difficulty, gets your creative juices flowing. Great if you already have a team ready, if not you will end up with the most insufferable teammates there are.
- lectures are pointless. don't like how russel has his toes out in sandals. read all the content, specifically docs written by other people (i.e. not russel's slides) most of it is common sense realistically. just make sure you properly follow the instructions. constantly check in with lecturer to make sure you're on track/ doing it right. try and

06-30214 06-30251	**Human-Computer Interaction** **Human-Computer Interaction (Extended)**	**20**	**-**

go into this if you know other people you work well with because it's a team project and randos in a team project is always a little rough. russel very much wants you to succeed and if you interact, it's very easy to. quizzes are piss. why was half of one just on eyes???? do i look like a med student to you? easy module to pass, moderate to get >70% in. is it worth your time? probably not but if you're just looking for credits, go for it. at least try though so your team isn't left picking up the slack for you

06-38964 06-38968	**Intelligent Data Analysis** **Intelligent Data Analysis (Extended)**	**20**	**-**

- Mid
- Only had Felipe on the module this year compared to having Peter as well last year
- Peter was very good at explaining and building up to the more complex topics last year, but that was not the case for Felipe - the first half of the content felt like a complete mess, but the second half which he has previously taught felt more coherent.
- IDA really needs a second lecturer, since Felipe often went on academic trips with a PGR student covering for him, but it didn't feel like it worked well
- Blehhhhhh. Got bait and switched by the excellent reviews last year - this was the first year that Felipe taught the first half of the module and it showed. The content on was vague, underspecified and hard to follow plus had minimal practical applications included. It was generally not great at getting you to understand anything, plus Felipe routinely went MIA and left the module to one of his TAs who didn't appear to especially enjoy teaching the module
- . The second half was far better and more polished, so it's not all awful, and Felipe's tutorials are definitely suited to what you need if the class makes that clear (eg. he did a whole one on teaching basic matrix maths). All of that said, some of my maths friends loved this module and got on fine with it so, if you're like me and are a bit hesitant about maths in general, beware. Otherwise, YMMV.
    - Oh also beware if you didn't do further maths at A-level because Felipe thinks either a) everyone did or b) the content in further maths is actually part of the regular course
    - He did seem to realise as the term went on that we weren't understanding and adjusted his teaching so may have a better idea of how to teach it next year?
- Wouldn't really recommend this module unless you are a maths fanatic. The paragraph above really nails what went wrong here. Also, the 2 hour long lectures can sometimes be so incredibly content heavy that my brain stopped working halfway through. This isn't because the content was especially difficult but because it was taught in quite a confusing way.
    - if you do take this module prepare yourself for the stress of Felipe screaming

06-38964 06-38968	**Intelligent Data Analysis** **Intelligent Data Analysis (Extended)**	**20**	**-**

down the mic all lecture

- I actually found Felipe's explanations to be very clear, moreso than any other lecturer. There's a lot of content in the first half, and it's very difficult maths, but i found his explanations focussed on conceptual understanding really helpful.
- I needed a refresh on A level Further maths for this module due to Felipe thinking it's part of the normal A level course, while I did find the first part confusing he was really helpful in office hours at explaining why things are done and where I might have gone wrong
- Awful slides, awful teaching, awful exam
    - The exam felt like it was meant to be open book, since Felipe talked quite a bit about his students in Mexico having open-book exams and doing well
- GET THEM BOYS.
- Ass exam wtf so happy I never have to look at IDA again
- HILARIOUS EXAM
    - (for all the wrong reasons, may I add)
- Horrible slides that are most the time hard to understand with tons of jargon
    - Not enough or appropriate practical questions to help support for the exam
        - Not enough is an understatement, there were none
    - The exam itself was diabolical a whole question worth 33 marks or something and left for interpretation, Felipe may be a lecturer of all time
    - Don't take this module 😭🙏 you're better off, thank god i don't need to see this module again
    - An additional note: Felipe said we didn't have to memorise any equations would have to derive some in the exam, in the exam NO EQUATIONS WERE GIVEN, it was all just a free for all in knowing what equation or formula or method to use
    - There was also alot of emphasis on Matlab in calculating algorithms like PCA rather than giving us actual easier maths calculations, it did absolute jack for exam questions and serves no other purpose than just teaching you matlab for the sake of it.
- Too much content, especially the first 5 weeks, not enough calculation examples. I feel that tutorials should be purely calculation examples not code. Exam marks were also unbalanced and it felt that there were too many questions. Overall, I would not choose this module again, I enjoyed the last 5 weeks but the first half is deadly and I'd say kind of boring in a way. I'm happy to be done with this

06-37809 06-37792	**Intelligent Interactive Systems** **Intelligent Interactive Systems (Extended)**	**-**	**20**

- No math or programming in this module. It discusses how to build systems that involve AI in a responsible way (Responsible AI), so things like explainability, privacy, and bias are discussed in this module. There are also topics from game theory such as prisoner dilemma and payoff matrix. We also discussed supervised

06-37809 06-37792	**Intelligent Interactive Systems** **Intelligent Interactive Systems (Extended)**	-	**20**

and unsupervised learning just like we did in AI 1 in the first year. In the exam, they could ask you about a specific problem and how you would solve it, in that case, you need to choose the correct approach (supervised or unsupervised) and state the algorithm that you will be using and why you discarded the alternatives and what are their weaknesses (why they do not apply to the problem). So they will not ask you about the algorithms themselves because this is not the module's scope. There are two assignments, the first is a 1500 literature review on bias in AI systems, and the second is designing an intelligent interactive system 1500 words as well. So far the marking is fair.

- resit was ok tho

06-30227 06-30244	**Intelligent Robotics** **Intelligent Robotics (Extended)**	**20**	-

  - This module has been completely reworked compared to previous years. It now involves no team coursework and no requirement to use ROS. The project basically requires you to implement one or a few of the algorithms presented in class to your own robotics project in a simulated environment and language of your choice. I found the content helpful to decide what to do for the project. The first part of the in person lectures involved quite a bit of maths that I struggled to understand, but luckily this didn't matter much as we were assessed on the coursework instead. The labs teach you the basics of ROS, and point you to some simulation packages, but for the most part you will need to learn all of the practical side yourself. They say it is recommended to use Linux for this module but I did the whole thing in Windows fine. I didn't get a lot of feedback from the proposal or the final report but the marking seems to be quite generous. Overall this module is a much more reasonable difficulty compared to previous years and I found the coursework to be challenging but fun.
- I would highly recommend this module to anyone who wants to get an insight into Intelligent Robotics. The module lead Melanie was by far the most supportive lead I have ever had. She helped with absolutely everything from project proposal to help with content. It is 100% coursework which I found beneficial, contrary to above I did receive a lot of feedback during the development phase of the project with both the coding and the write up. The above does mention the use of ROS which is heavily emphasised however you can use WeBots and still be fine however it is not taught at any point so you will have to learn how to use it. In addition, the use of Linux is highly recommended, however I used MacOS and still managed fine. Melanie was very helpful with helping get to grips with WeBots. One thing I would mention is that the specification for the coursework is essentially to develop a simulated robot which displays some intelligent behaviour what I decided to do revolved around the early weeks which meant the majority of the lectures were completely redundant to me, which depending how you see it is either a positive or negative.

06-40090 06-40095	**Intelligent Software Engineering** **Intelligent Software Engineering (Extended)**	**20**	**-**

- A module with promise.
    - Weeks 1-5 looked at how AI is used to solve problems (e.g. classifying bug reports). Generally, explains the background and techniques of research (interesting to see how the techniques of AI1/AI2 are actually used in research).
    - Weeks 7-10 is a more holistic view of AI in software (e.g. AI music generation).
    - Coursework - Pick one of two tasks. Build your own “AI to solve problem”, and beat a baseline (the problems they suggest are the content from weeks 1-5 - I would recommend using one of these) OR write a literature review.
        - Coursework is released early in the module, and due at the end - so lots of time.
        - Having two options for coursework is great
        - The tool building was fun (but if you haven’t had any experience with implementing AI, for example PyTorch, you will have to self learn this)Beware of literature review fatigue (especially if you are doing DDS, IIS or FYP)
    - Lectures generally have **lots** of content, with no breaks. It has been hard to tell what will be in the exam (although we have recently had detailed
    - information on what slides could come up in the exam - about 200 pages total).
    - Tao is friendly, and replies to emails quickly.
- Hard to give opinion as no marks have been returned yet
- Great coursework if you like making AI, horrible content and exam

06-38965 06-38969	**Machine Learning** **Machine Learning (Extended)**	**20**	**-**

- **If you fall behind by even a week and don't catch up you will die**
- Leandro is a very good lecturer, I found the content taught by him pretty difficult even though AI 1 all made sense (which he also taught me) but that was probably a skill issue because he definitely did his best to go through complicated concepts slowly and thoroughly. Jian on the other hand… the second half of this module was just being bombarded by miscellaneous facts and random equations that I felt had not even a hint of coherence with each other. I would say the second half had simpler content but it's hard to judge when the teaching was this bad for it.
- Leandro is an excellent lecturer and manages to break down the content well such that, although very difficult, it can be understood if you put the time in. As someone else said, you can not breeze through this module and it's essential you understand each concept well before moving onto the next topics. If you are someone that does not like maths a lot do not even bother, it's the most maths heavy module I've enrolled in and the concepts are difficult to visualise without having a good understanding of linear algebra. It is however very rewarding once you finally manage to make sense of it. Second half of the module is slightly lighter on the maths but the lecturer is also not as good as Leandro.
- This module felt like a proper continuation of AI. honestly, AI2 wasn't that great for me. The module is split into two very different halves. The first half is taught by Leandro, and he's a great lecturer. In all of his lectures, I understood everything except maybe one or two slides, which I could easily review at home and watch Leandro recording again without needing YouTube or other resources. This part mainly covers SVMs, but the first two weeks are just an introduction, gradient descent, and logistic regression. Each week builds on the previous one, which is nice because it feels like you're studying one continuous topic. However, if you fall behind, it can be tough to catch up. The second half, though, is a nightmare both in terms of the topics and the lecturer's explanations. The content is much more theoretical, covering things like learning feasibility, bias and variance, and Hoeffding's inequality. These aren't algorithms but rather theoretical concepts that can be tricky to grasp. you can find these topics on YouTube in the Caltech Learning From Data playlist. If the module only included the first half, I'd recommend it easily. But the second half makes it harder to enjoy—not just because the content is difficult, but also because the lecturer just reads off the slides without explaining things clearly. That said, it's still a solid module, and it covers important core concepts
- Hard but very interesting module, it is difficult to keep up with at times, but Leandro is an excellent teacher and explains things really well. Second half is not taught as well, but it has easier content. Very mathematical module, would really recommend it for maths and comp sci students.
- Very mathematical, yet practically useless module. You get taught the same basic foundation (Overfitting and underfitting, linear regression and logistic regression) again but with more maths. Literally useless for modern AI topics, pick neural

06-38965 06-38969	**Machine Learning** **Machine Learning (Extended)**	**20**	**-**

computation instead.

- idk. i left the first lecture halfway through because the smell in the room was atrocious and only ever went back again once. please shower if you take this module. content very similar to ai1 and ai2. some other interesting bits and pieces in there. if you fall behind, just drop out. not very interesting, be prepared to see 50 million slides on gradient descent. if you're interested in ai and want to take this as a basis for others modules. save your time and energy and just do the others, use your ai1 and 2 knowledge and look back at old notes. if you just want to take an ai module to refresh your memory, go for it. if you want to take an ai module for funsies, do neural comp instead

06-29289 06-25689	**Mobile & Ubiquitous Computing** **Mobile & Ubiquitous Computing (Extended)**	**-**	**20**

- If you like dealing with out of date libraries and dead code, this is the module for you!
- I do not recommend this module. Last year they complained about it being too easy and now they have made it hard and a lot of work. The team assignment is 15% but should be much more.
  If you email them they don't respond (Chris doesn't anyways)
  The lecturers are so lost and useless I feel like they don't even know what's going on half the time. If you ask them a question they do not know. There was a part of the lecture on resistors and batteries and stuff, everything he was saying was incorrect. The slides were correct but his explanation on the logic wasn't. ChatGPT is teaching me everything. I'm

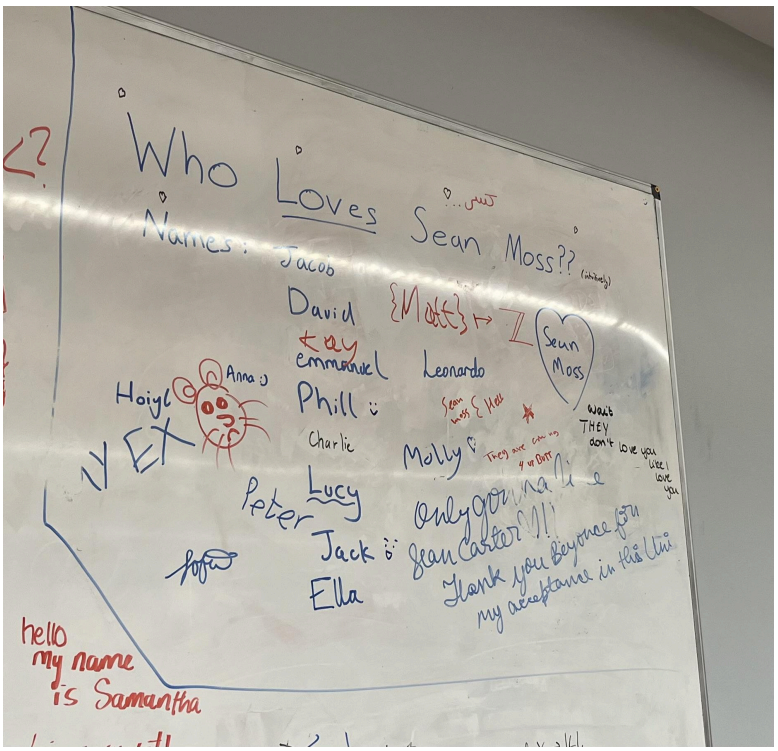
06-32167 06-32212	**Neural Computation** **Neural Computation (Extended)**	**20**	**-**

- Concepts were explained clearly and if you have a general interest in AI I would actually recommend this module. The slides contain all the information that is required for the assessments without any ambiguity and are well annotated. Of the two lecturers, Alexander Krull is far far better and you can tell he has a genuine passion for teaching the topic and tries to emphasise the harder concepts and break them down for you. Some of the content is challenging but I believe the slides do a good job of explaining them clearly. I would highly recommend using YouTube videos by 3Blue1Brown to clarify some of the foundational concepts
    - I second the 3B1B recommendation!!
- If I had any other decent option available to me I would have left this module behind in Week 1. Genuinely up there as one of the worst modules I've done. The coursework section was two 10 question quizzes that weren't too difficult if you'd properly gone over the content, and were open book, so that was pretty fair. But dear god, motivating myself to go to these lectures was like pulling teeth. They were dry as all hell and had my brain melting out of my ears with boredom. I would have rather done CVI again. As someone who is lukewarm on AI, pick a different module and save your brain.
- If you want to understand new AI topics like transformers and how ChatGPT works, this is a great course. The topics covered include RNNs, autoencoders, diffusion models, GANs, and transformers. I personally didn't like the Edgbaston lectures as they were difficult to understand, so I watched the Dubai recordings instead, he explains everything really clearly (Kashif Rajpoot), he's the best. Each week, there's a lab file in Jupyter Notebook. Some quiz questions are based on these labs. Most of the code is already written, but there are a few lines left for you to complete nothing too complicated, however the labs could be very long. There are two online quizzes with 10 multiple-choice questions. They aren't too hard if you've studied well, but make sure to read the questions carefully. If this dubai lecturer (Kashif) was not there, I would've had a completely different opinion about this module.
- Incredible for the new AI topics and implementing them in the weekly labs. Not as theoretical as ML but more practical. Do this if you actually want to learn the real architecture behind ai instead being another "prompt engineer"

06-30230 06-30256	**Programming Language Principles, Design, and Implementation** **Programming Language Principles, Design, and Implementation (Extended)**	**20**	**-**

- This is an extremely challenging module, but it is also very rewarding.
- Sean and Vincent are great lecturers
    - 100% agree, they're both fantastic

06-30230	**Programming Language Principles, Design, and Implementation**	20	-
06-30256	**Programming Language Principles, Design, and Implementation (Extended)**		



- Sean teaches the first half, which is Haskell, lambda calculus, types; Vincent teaches the second half, which is compiler structure
- This module is effectively Theories of Computation - Part 2 (also see: Computer Science Theory - The Module)
- One of my favourite modules of all time (I am 100% biased as my FYP is just this module)
- I really enjoyed this module, though the second half (on compilers) can get a little repetitive as each section was just algorithm after algorithm. This was taught very well by Vincent regardless.
- “We now extend the lambda calculus … “ x 100
- Lambda calculus very hard, i am not him but also taught well. Minor skill issue on my end tbh. Second half is more ‘actual’ computer things and not what feel like some made up ancient hieroglyphs.
- Lambda calculus content is very reminiscent of first year logic; you do lots of tree-based proofs for operational semantics/type checking (albeit IMO the proof systems are more straightforward than natural deduction as it’s always clear what inference rules to apply). If you enjoyed MLFCS and ToC, I couldn’t recommend this module highly enough.
- Content is very challenging (except for the first 2 weeks, don't be lulled into a false sense of security) but I found it really interesting. Attempting all the questions on the weekly problem sheets is of utmost importance as that is the only way to learn the

06-30230 06-30256	**Programming Language Principles, Design, and Implementation** **Programming Language Principles, Design, and Implementation (Extended)**	**20**	**-**

content, because despite the lectures being taught well, you will come out of many having understood barely anything. Take this module if you're interested in theory but avoid if you're not willing to work hard.

- This is the best module I have taken. Vincent and Sean are excellent lecturers, the slides and their explanations are super clear and they are always happy to answer your questions and make time to re-explain things. The content can be challenging but if you are interested in theory, programming languages or doing loads of proof trees this is the module for you. Also! There's so many problem sheets so you get loads of practice and example questions which help so much for the assignments. 100000/10. (also, theres not a lot of assumed knowledge for this, they really do teach a lot from scratch which was very helpful)
- Only take if your have an interest in theory, not saying you can't do well if you don't like i did you will be constantly annoyed by the question "how dose this apply to anything remotely modern day." also it's applicability to anything modern day may be a bit questionable due to the age of the content taught. Thats mostly the first half the second half is much better.
    - I don't agree with this myself; lambda calculus is the foundation and a strong influence behind modern programming languages, so it is everywhere in modern day applications. Also, this is clearly a theoretical module, so the question of "where does this apply to real life/the modern world" isn't what such a module tries to answer; theory may build on older ideas, but they're still very much applicable in modern research, development, and the world of work, and the future of these too
- I will be a black sheep and argue that this isn't crazy challenging - a lot of the module content can sound a bit weird and unfamiliar (especially the first half w/ lambda calculus and type stuff) but ultimately, a lot of what you actually have to do for the module is just following algorithms and rules. \Gb-reducing lambda-terms, type-checking, making and using push-down automata… it's just learning the sequence of steps you have to follow and then doing it. They also provide a sheet in the exam with a lot of the rules so you don't even have to learn them all. If anything, I think that made some of the module rather boring to me, but it helps in the difficulty.
- I also agree with the above in terms of it's not that hard once you get enough practice so that you're familiar with it (by saying that I might have jinxed myself into failing but eh), though imo it's really interesting and not at all boring and the teaching was great

06-30231 06-30257	**Security of Real-World Systems** **Security of Real-World Systems (Extended)**	**20**	**-**

- Best module I've done at this uni, great lecturers, good content and assignments, brilliant overall.

06-30231 06-30257	**Security of Real-World Systems** **Security of Real-World Systems (Extended)**	**20**	**-**

- I don't like being overdramatic but holy shit this changed the way I looked at cyber security or tech in general
- The assignments were better balanced for workload this year, so 20% of the module mark felt like 20% rather than 100%
- Tom's lectures and content were absolutely amazing and when I thought they couldn't get better David came along and blew everyone's mind. The lecturers, the content, the assignments (I found them really fun personally) were all wonderful! 10/10 You should take this
- Could not recommend a module more than this one. Unless you absolutely HATED s+n (like failed it hated bc you couldn't stand it) then you MUSt take this module. Lecturers are great, assignments are really cool, will change the way you look at cybersecurity. 10/10 would recommend. You also get to hack a game. It sounded like it was a bit dodgy for mac people (just dont use a mac lmao - coming from someone with a mac) but hopefully they will fix that for next year since the game hacking is new. Also needed to use a vm/wsl if you weren't on linux (i did it through docker - again a mac issue) tldr don't use a mac bc it overcomplicates everything
    - The game hacking isn't new but it was a team project last year, which has been changed now - "🤓erm actually the game hacking isn't new"
- I have nothing to add to the above apart from 👍😀👍 take this module, I have no regrets
- Genuinely fun module, the assignments were great And did actually reinforce what we'd been taught.
- Very well taught
- Great module, the assignments were actually fun 🤯. Decently well taught I think but tbh I have forgotten all the material anyway so.
- By far the best module I have had at uni so far, incredible lectures and assignments, and might be the first module ever where I am looking forward to the revision lecture next semester.
- I actually liked the assignments and wanted to do them. 10/10 would recommend. You may need access to an x86 system for some of the assignments though, Mac users beware
    - This isn't true you can do them on mac but is more effort due to mac being bad
- If you liked Security and Networks I recommend picking this module, the assignments are a cake walk, and the lecturers (Tom Chothia, David Oswald) are really passionate about what they do so every lecture feels like you're watching a random youtube documentary you stumble upon at 3AM in the morning when you know you should be sleeping for a lecture you have at 9AM.

06-34240	**Secure Software and Hardware Systems**	**-**	**20**

- Postgraduate Masters only module.
- A lot of the content is very similar to Real World Security, but I would say this is a lot easier. There are two group assignments which are fairly straightforward. Make sure

06-34240	**Secure Software and Hardware Systems**	**-**	**20**

you get to know other people taking the module so you can find a good group. A nice easy module to take so I can lock in on project.

-

06-30208	**Teaching Computer Science in Schools**	**20**
06-30235	**Teaching Computer Science in Schools (Extended)**	

- Lectures are basically yap sessions and you learn (close to) nothing. Expect semester 1 to be ***dry*.** Being able to find a school to do your 8 days of placement in early on will take a lot of pressure off you.
- Useful to know: being accepted onto the module isn't a confirmation you will get a placement! Bob used first semester lectures to check in with if anyone is able to organise their own and prioritise people for placements with good attendance. He will try his best to get as many placements sorted as possible but don't expect to know your placement before Christmas. You will get an email before Christmas to tell you how to get your DBS with uni.
- Module lead has an average response time of 2 working weeks and the taught part of the module taught me nothing. Having said that it's an amazing experience teaching with a friend building skills you can not really get through any of the other modules.

Further Advice

For Everyone

- If you're planning on doing a 3:1 split then be warned that it is very challenging in the first semester, *especially* if you're picking difficult modules like PLPDI. It might be better now than in previous years given that all exams are in summer, but remember that you also have your project this year and it is very easy to fall behind.
    - I also wouldn't recommend doing the inverse (1:3 split) unless you're confident you can lock in on your project in semester one and basically finish it.
- If you're struggling to pick between modules, remember you've got the option of adding a year and doing an MSci! 9 modules baybeeeee
    - Swapping onto MSci has saved my fyp, I actually have an idea of what to do now thanks to taking PLPDI, so if you are struggling with fyp ideas I do recommend MSci
    - When in doubt, do a panic masters
    - Unfortunately this did not work for me lmao I might be cooked x4
- don't do a panic masters. i did. got to third year and my god i wish i was graduating this year. only do a panic masters if you got nothing else going for you. stay in academia forever!!!

For Maths and Computer Science Students

- Maths + cs students aren't real

For Degree Apprentice Students

- DA requirements can be found here.
- Remember to reset your password REGULARLY or you will need to travel into the nearest office to reset it